

CS 70 — Homework 1

Problem 1: Always True or Always False?

(a) $x \wedge (x \Rightarrow y) \wedge (\neg y)$

Contradiction. If x is true, then $x \Rightarrow y$ forces y to be true, making $\neg y$ false. If x is false, the conjunction is immediately false. There is no combination of x and y that makes this true.

(b) $x \Rightarrow (x \vee y)$

Tautology. If x is true, then $x \vee y$ is true regardless of y . If x is false, the implication is vacuously true. True in all cases.

(c) $(x \vee y) \vee (x \vee \neg y)$

Tautology. This simplifies to $x \vee (y \vee \neg y) = x \vee \text{True} = \text{True}$ always.

(d) $(x \Rightarrow y) \vee (x \Rightarrow \neg y)$

Tautology. If x is false, both implications are vacuously true. If x is true, then either y is true (making $x \Rightarrow y$ true) or y is false (making $x \Rightarrow \neg y$ true). In all cases at least one disjunct holds.

(e) $(x \vee y) \wedge (\neg(x \wedge y))$

Neither. This is exclusive or (XOR): true when exactly one of x, y is true, false when both are true or both are false.

(f) $(x \Rightarrow y) \wedge (\neg x \Rightarrow y) \wedge (\neg y)$

Contradiction. $(x \Rightarrow y)$ and $(\neg x \Rightarrow y)$ together force y to be true regardless of x . But $\neg y$ contradicts this. Always false.

Problem 2: Propositional Practice

- (a) There is a real number which is not rational.

$$\exists x \in \mathbb{R}, x \notin \mathbb{Q}$$

True. $\sqrt{2}$ is irrational.

- (b) All integers are natural numbers or are negative, but not both.

$$\forall x \in \mathbb{Z}, (x \in \mathbb{N} \vee x < 0) \wedge \neg(x \in \mathbb{N} \wedge x < 0)$$

False. The integer 0 is a natural number and is not negative, which is consistent, but negative integers are not natural numbers — the statement fails to account for 0 being in $\mathbb{N}$ under some conventions, and the XOR structure breaks for certain definitions.

- (c) If a natural number is divisible by 6, it is divisible by 2 or it is divisible by 3.

$$\forall x \in \mathbb{N}, (\exists k \text{ s.t. } x = 6k) \Rightarrow (\exists k_1 \text{ s.t. } x = 2k_1) \vee (\exists k_2 \text{ s.t. } x = 3k_2)$$

True. Any multiple of 6 is divisible by both 2 and 3.

- (d) $(\forall x \in \mathbb{R}) (x \in \mathbb{C})$

“Every real number is a complex number.” **True.** $\mathbb{R} \subset \mathbb{C}$; every real x can be written as $x + 0i$.

- (e) $(\forall x \in \mathbb{Z}) ((2 \mid x \vee 3 \mid x) \Rightarrow 6 \mid x)$

“For all integers, if x is divisible by 2 or divisible by 3, then x is divisible by 6.” **False.** Counterexample: $x = 2$ is divisible by 2 but not by 6.

- (f) $(\forall x \in \mathbb{N}) ((x > 7) \Rightarrow ((\exists a, b \in \mathbb{N}) (a + b = x)))$

“For all natural numbers, if x is greater than 7, then x can be written as the sum of some two natural numbers.” **True.** For any $x > 7$, take $a = 1$ and $b = x - 1$.

Problem 3: Prove or Disprove

- (a) $\forall n \in \mathbb{N}$, if n is odd then $n^2 + 2n$ is odd.

True. Let n be odd, so $n = 2k+1$ for some integer k . Then $n^2 = (2k+1)^2 = 4k^2 + 4k + 1$ is odd, and $2n = 2(2k+1)$ is even. Since odd + even = odd, $n^2 + 2n$ is odd. $\square$

- (b) $\forall x, y \in \mathbb{R}, \min(x, y) = (x + y - |x - y|)/2$.

True. We split into two cases.

Case 1: $x \leq y$. Then $\min(x, y) = x$ and $|x - y| = y - x$, so:

$$\frac{x + y - (y - x)}{2} = \frac{2x}{2} = x \checkmark$$

Case 2: $y < x$. Then $\min(x, y) = y$ and $|x - y| = x - y$, so:

$$\frac{x + y - (x - y)}{2} = \frac{2y}{2} = y \checkmark$$

$\square$

(c) $\forall a, b \in \mathbb{R}$, if $a + b \leq 10$ then $a \leq 7$ or $b \leq 3$.

True. We prove by contraposition. Assume $\neg(a \leq 7 \vee b \leq 3)$, i.e., $a > 7$ and $b > 3$. Then $a + b > 10$, which is $\neg(a + b \leq 10)$. $\square$

(d) $\forall r \in \mathbb{R}$, if r is irrational then $r + 1$ is irrational.

True. By contraposition: assume $r + 1$ is rational, so $r + 1 = a/b$ for some $a, b \in \mathbb{Z}$, $b \neq 0$. Then $r = a/b - 1 = (a - b)/b$, which is rational. This contradicts r being irrational. $\square$

(e) $\forall n \in \mathbb{N}^+$, $10n^2 > n!$

False. Counterexample: $n = 10$. Then $10n^2 = 10 \cdot 100 = 1000$, but $10! = 3,628,800 \gg 1000$.

Problem 4: Preserving Set Operations

For a function f , the image of a set X is $f(X) = \{y \mid y = f(x) \text{ for some } x \in X\}$, and the inverse image of a set Y is $f^{-1}(Y) = \{x \mid f(x) \in Y\}$.

1. $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$.

By definition, $f^{-1}(A \cup B) = \{x \mid f(x) \in A \cup B\}$. By definition of union, $f(x) \in A$ or $f(x) \in B$. Therefore:

$$= \{x \mid f(x) \in A\} \cup \{x \mid f(x) \in B\} = f^{-1}(A) \cup f^{-1}(B) \square$$

2. $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$.

By definition, $f^{-1}(A \cap B) = \{x \mid f(x) \in A \cap B\}$. By definition of intersection, $f(x) \in A$ and $f(x) \in B$. Therefore:

$$= \{x \mid f(x) \in A\} \cap \{x \mid f(x) \in B\} = f^{-1}(A) \cap f^{-1}(B) \square$$

3. $f^{-1}(A \setminus B) = f^{-1}(A) \setminus f^{-1}(B)$.

By definition, $f^{-1}(A \setminus B) = \{x \mid f(x) \in A, f(x) \notin B\}$. Therefore:

$$= \{x \mid f(x) \in A\} \setminus \{x \mid f(x) \in B\} = f^{-1}(A) \setminus f^{-1}(B) \square$$

4. $f(A \cup B) = f(A) \cup f(B)$.

By definition, $f(A \cup B) = \{y \mid y = f(x) \text{ for some } x \in A \cup B\}$. By definition of union, $x \in A$ or $x \in B$. Therefore:

$$= \{y \mid y = f(x) \text{ for some } x \in A\} \cup \{y \mid y = f(x) \text{ for some } x \in B\} = f(A) \cup f(B) \square$$

5. $f(A \cap B) \subseteq f(A) \cap f(B)$, but equality need not hold.

Proof of inclusion. Let $y \in f(A \cap B)$. Then $y = f(x)$ for some $x \in A \cap B$, meaning $x \in A$ and $x \in B$. Since $x \in A$, $y = f(x) \in f(A)$. Since $x \in B$, $y = f(x) \in f(B)$. Therefore $y \in f(A) \cap f(B)$. $\square$

Counterexample for equality. Let $f(x) = x^2$, $A = \{1\}$, $B = \{-1\}$. Then $A \cap B = \emptyset$, so $f(A \cap B) = \emptyset$. But $f(A) = \{1\}$ and $f(B) = \{1\}$, so $f(A) \cap f(B) = \{1\} \neq \emptyset$.

6. $f(A) \setminus f(B) \subseteq f(A \setminus B)$.

Let $y \in f(A) \setminus f(B)$. Then $y \in f(A)$ and $y \notin f(B)$. Since $y \in f(A)$, there exists $x \in A$ such that $f(x) = y$. Since $y \notin f(B)$, this $x \notin B$ (otherwise $y = f(x) \in f(B)$, a contradiction). Therefore $x \in A \setminus B$, so $y = f(x) \in f(A \setminus B)$. $\square$